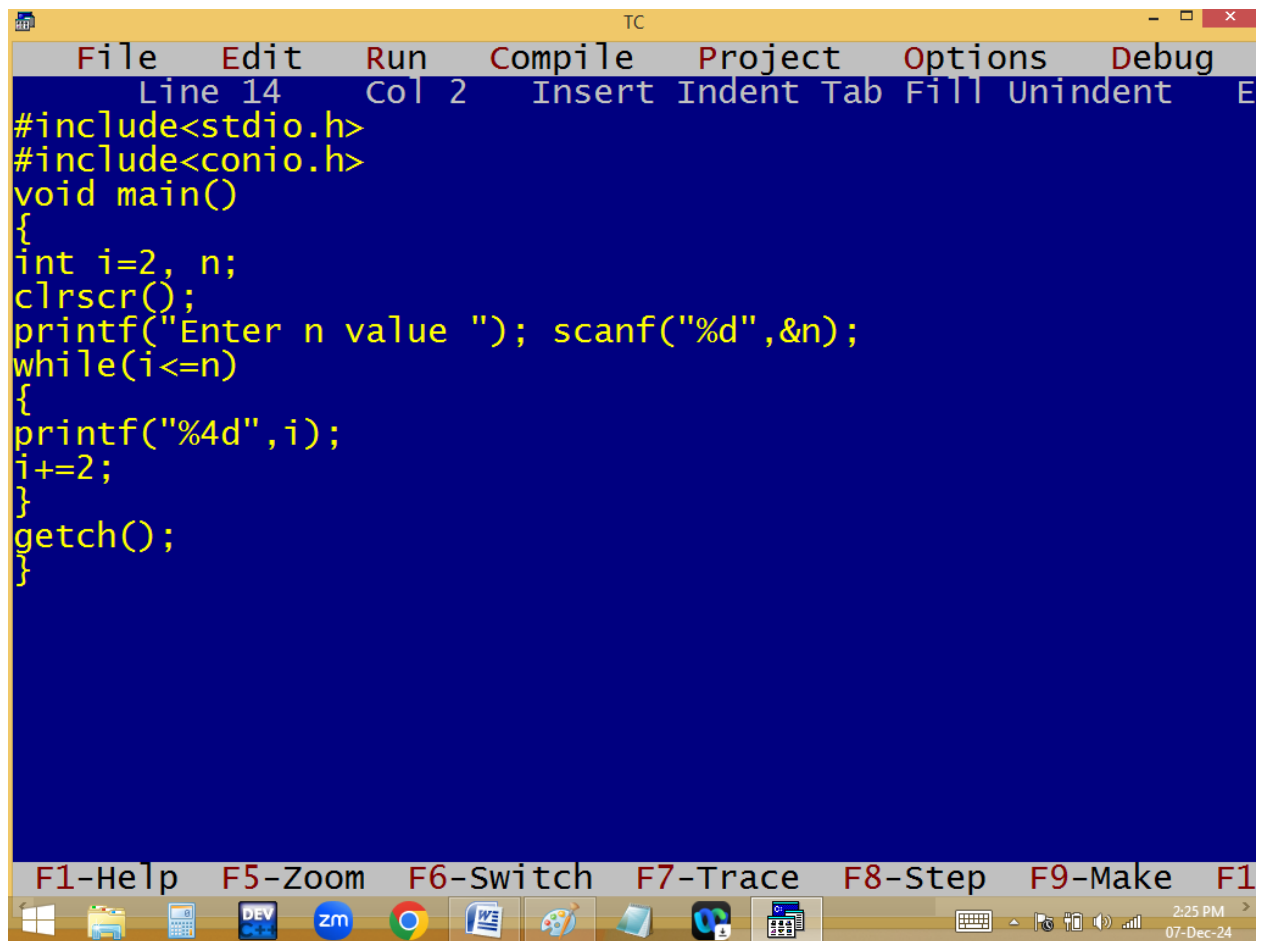
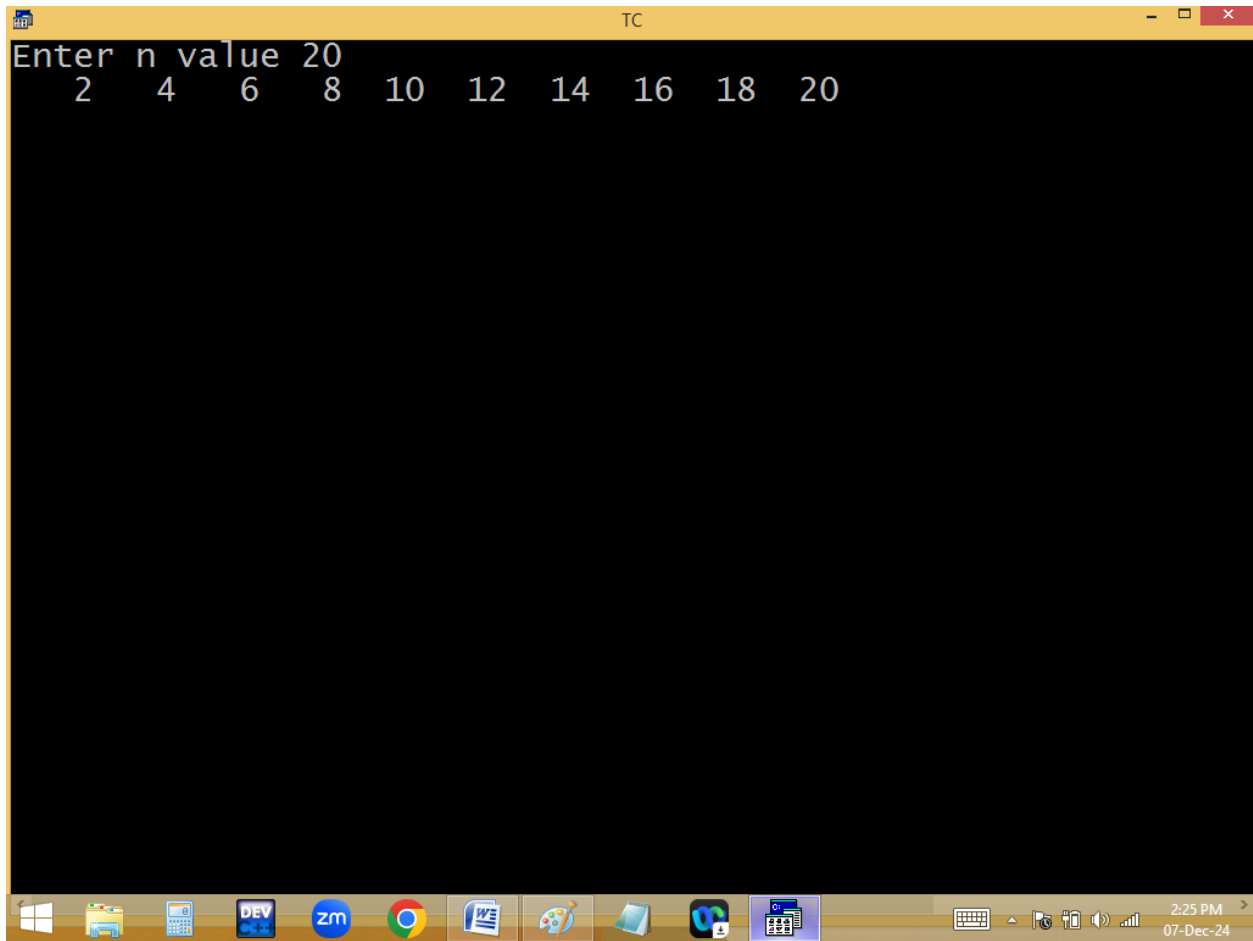


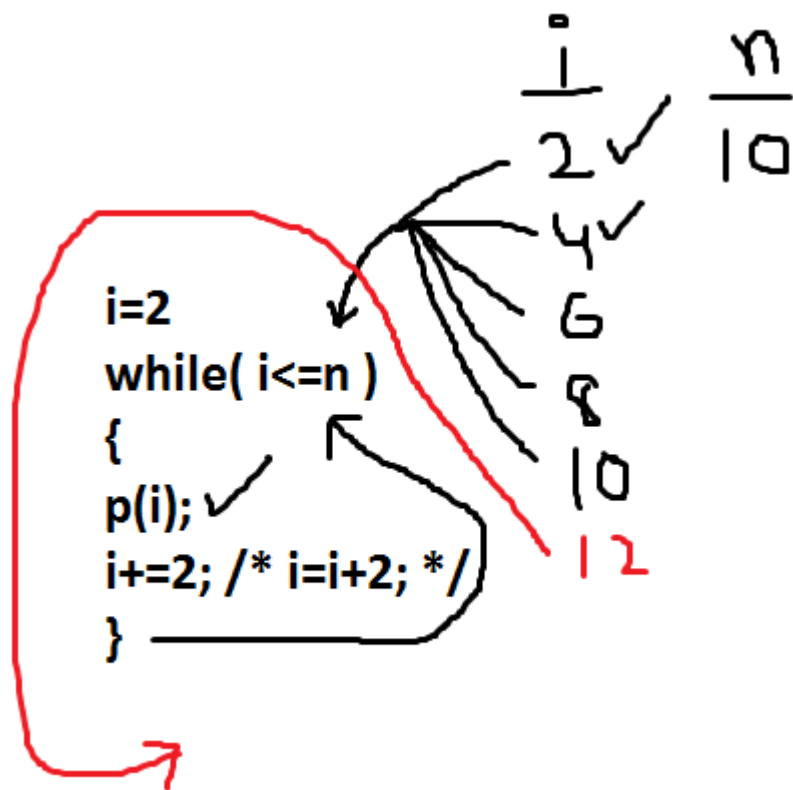
Printing 1..n even no's:

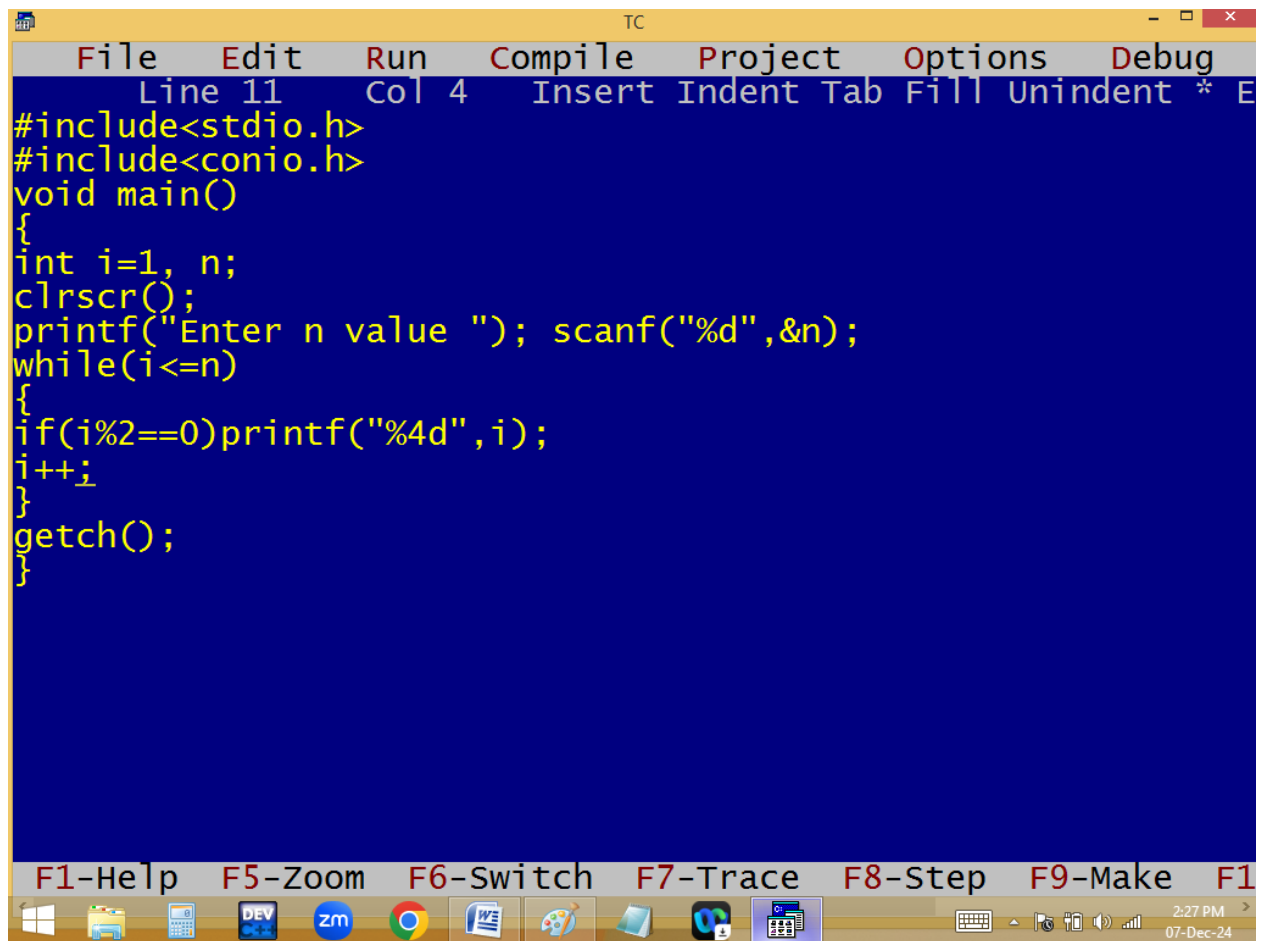


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 14 Col 2" and lists editing options: Insert, Indent, Tab, Fill, Unindent, and End. The code is written in yellow text on a dark blue background. It includes headers for stdio.h and conio.h, defines a main function, declares variables i and n, clears the screen, prompts the user for a value of n, and uses a while loop to print even numbers from 2 to n. The bottom status bar shows function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the bottom contains icons for various applications and the system clock shows 2:25 PM on 07-Dec-24.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=2, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
printf("%4d",i);
i+=2;
}
getch();
}
```



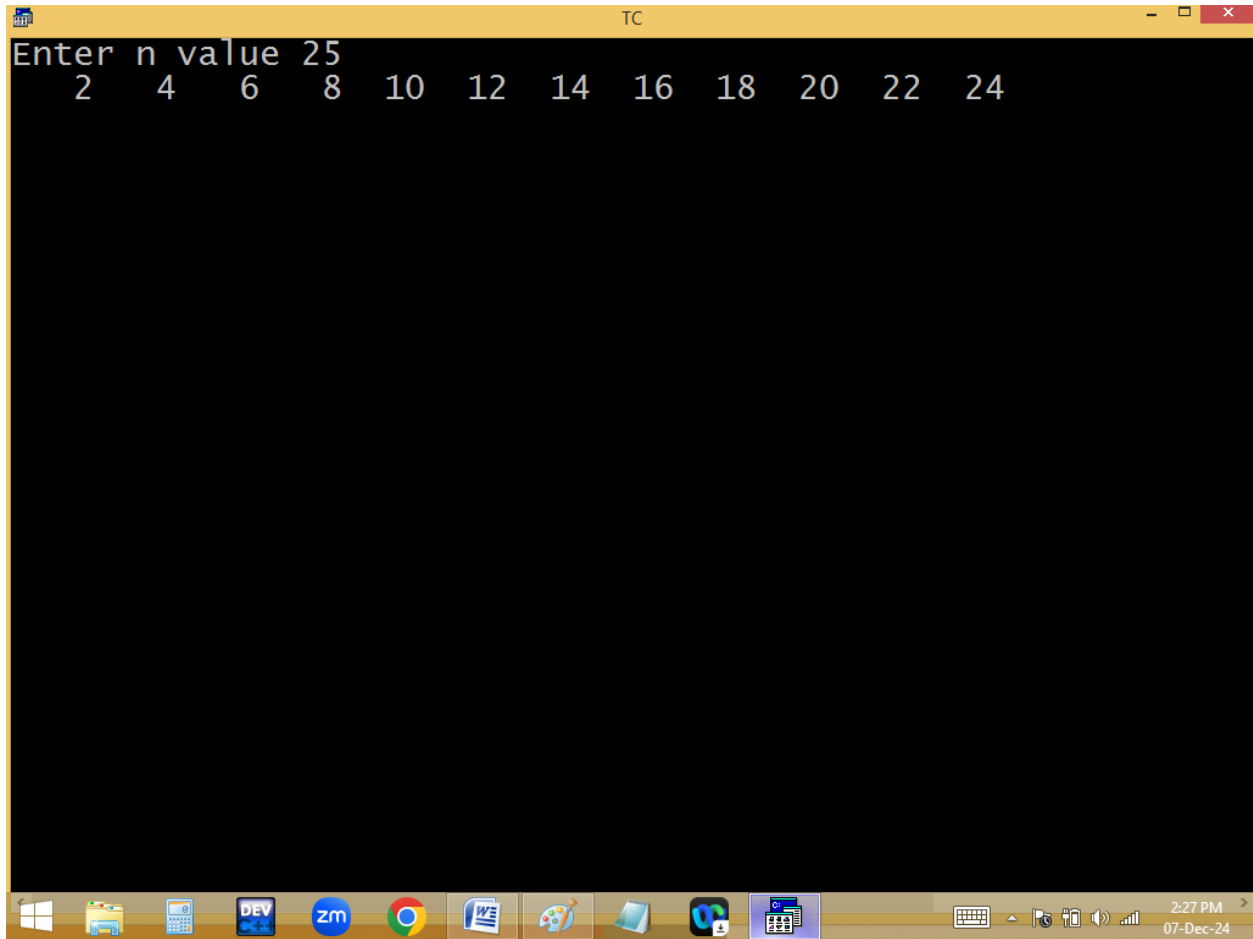




The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, there is a submenu with options: Line 11, Col 4, Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i=1, n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    while(i<=n)
    {
        if(i%2==0)printf("%4d",i);
        i++;
    }
    getch();
}
```

At the bottom of the window, there is a status bar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F1. Below the status bar is a Windows taskbar with various icons including the Start button, File Explorer, Calculator, DEV, zm, Google Chrome, Word, Paint, and a folder icon. The system clock in the bottom right corner shows the time as 2:27 PM and the date as 07-Dec-24.



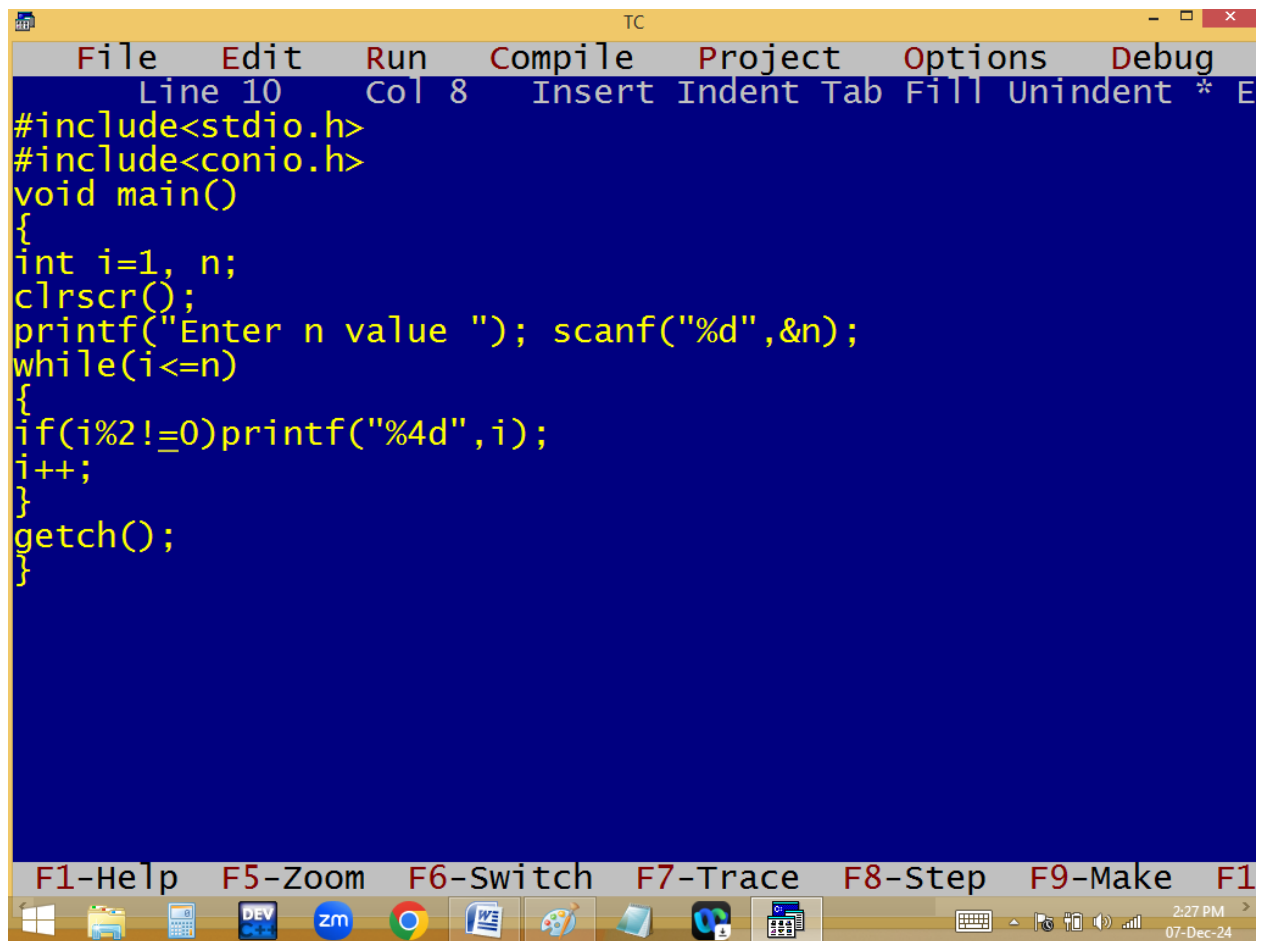
```

i=1;
while(i<=10)
{
  if(i%2==0)p(i);
  i++;
}

```

1
 1 x
 2 ✓
 3 x
 4 ✓
 -
 -
 10

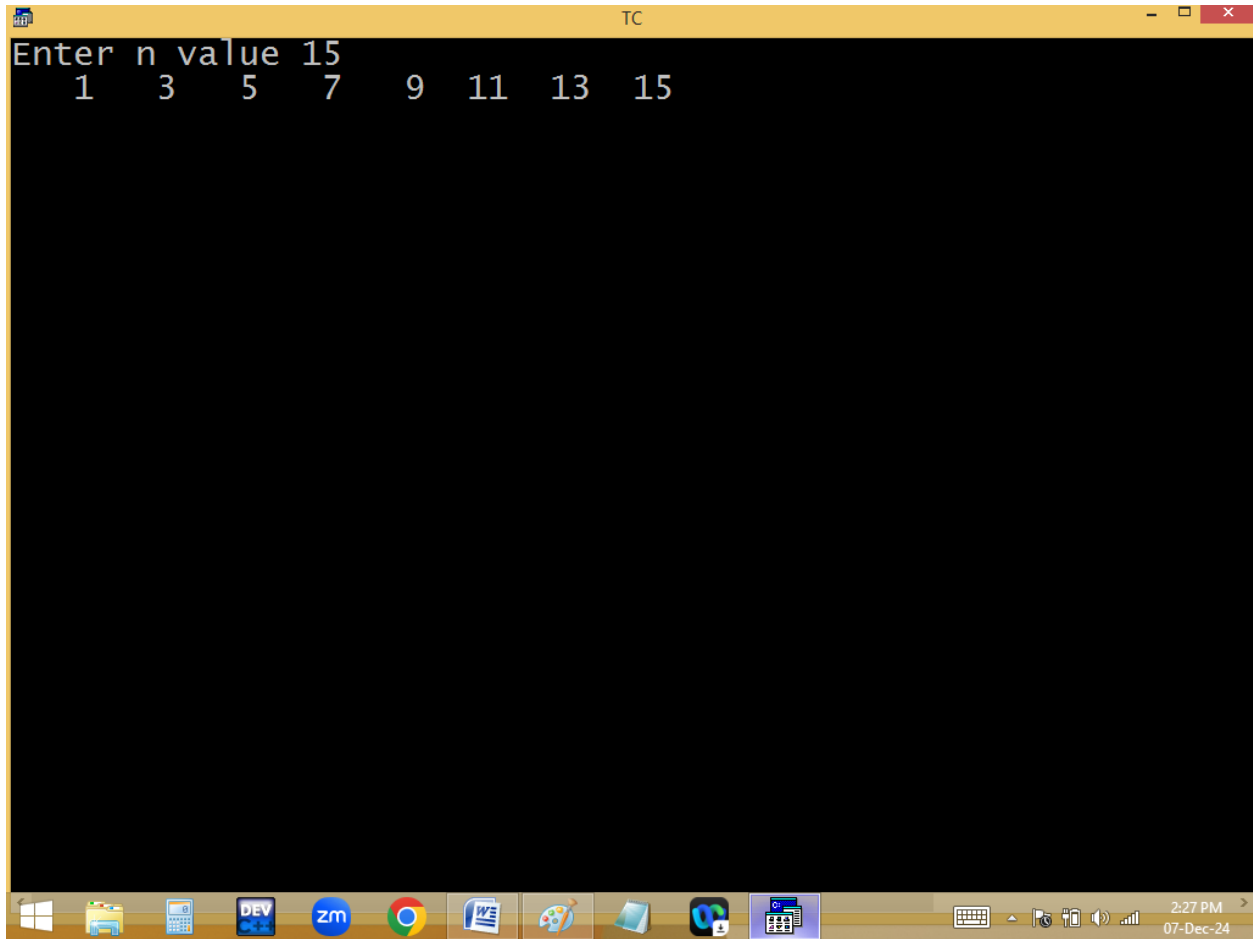
Odd no's:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10 Col 8" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i=1, n;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    while(i<=n)
    {
        if(i%2!=0)printf("%4d",i);
        i++;
    }
    getch();
}
```

The bottom of the window features a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "2:27 PM" and "07-Dec-24".



```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 5 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
printf("%4d",i);
i+=2;
}
getch();
}
```

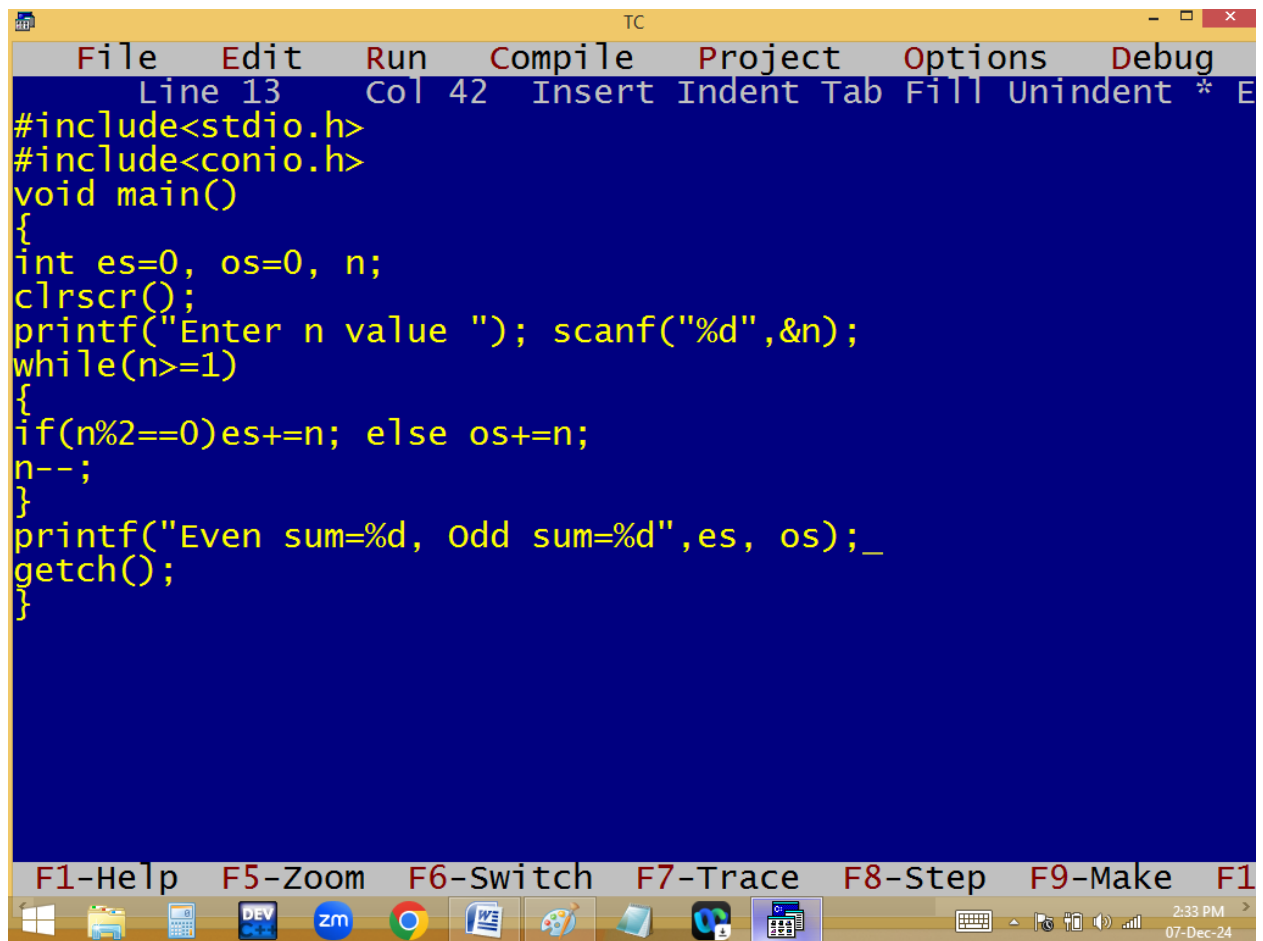
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:28 PM 07-Dec-24

```
TC
Enter n value 10
1 3 5 7 9_
```

Print 1..n even, odd no's sum:

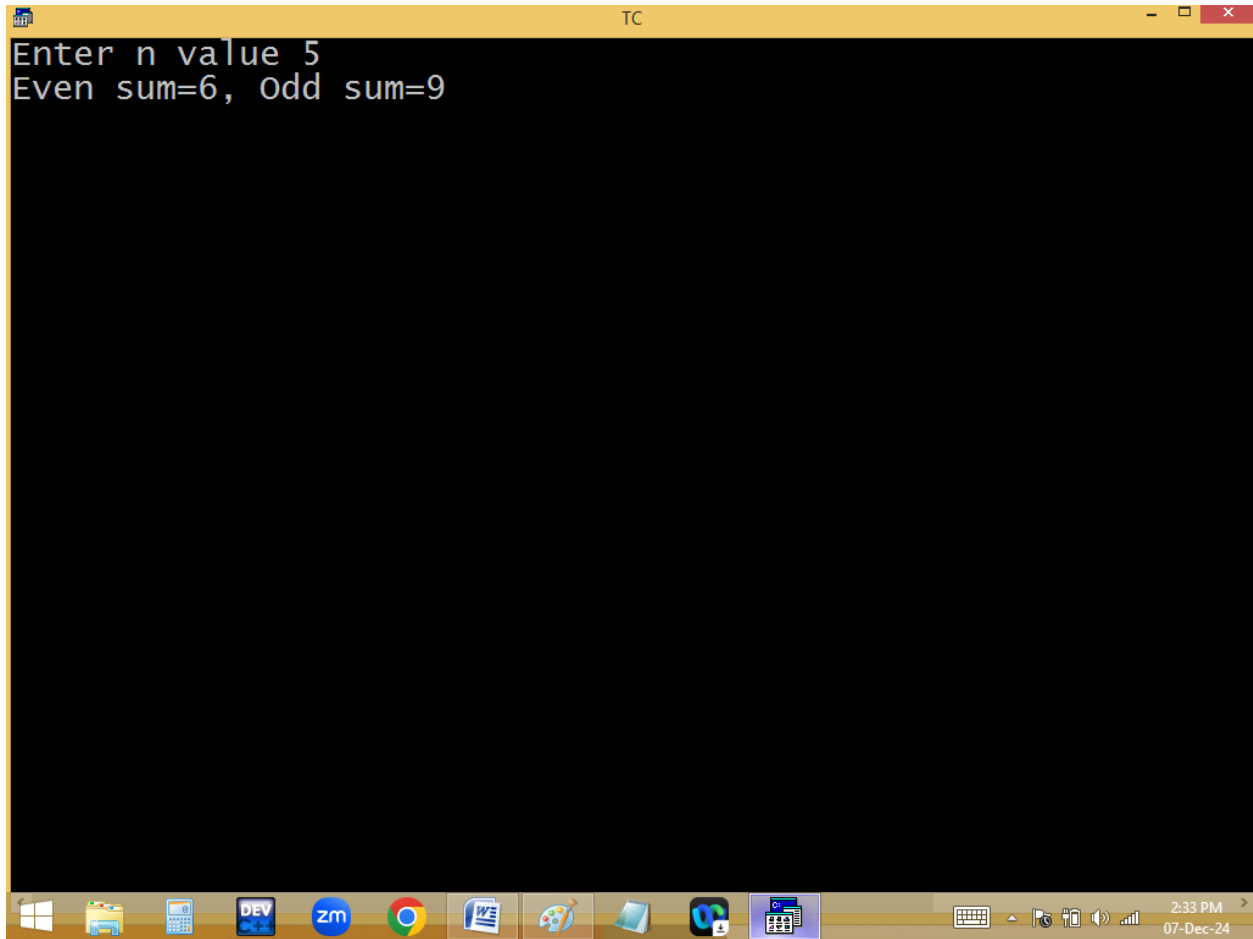
$$n = 5 \begin{cases} 1 + 3 + 5 = 9 \\ 2 + 4 = 6 \end{cases}$$

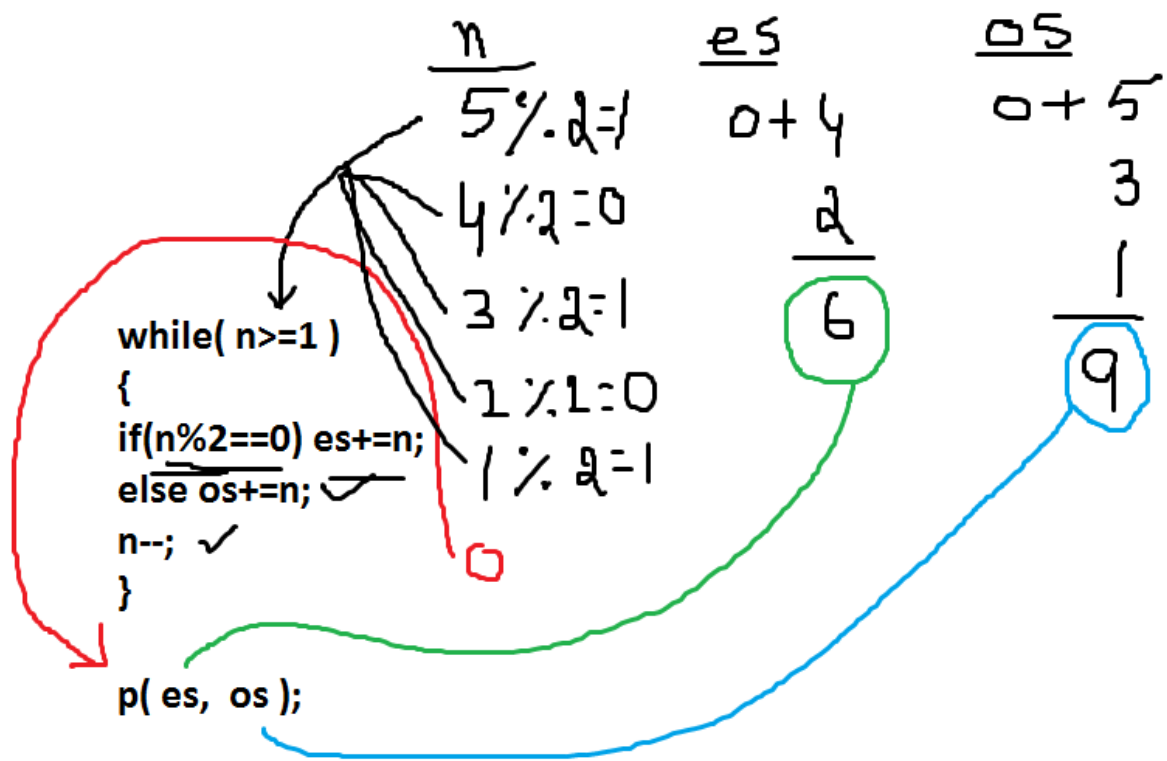


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 42", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a blue background and contains the following C code:

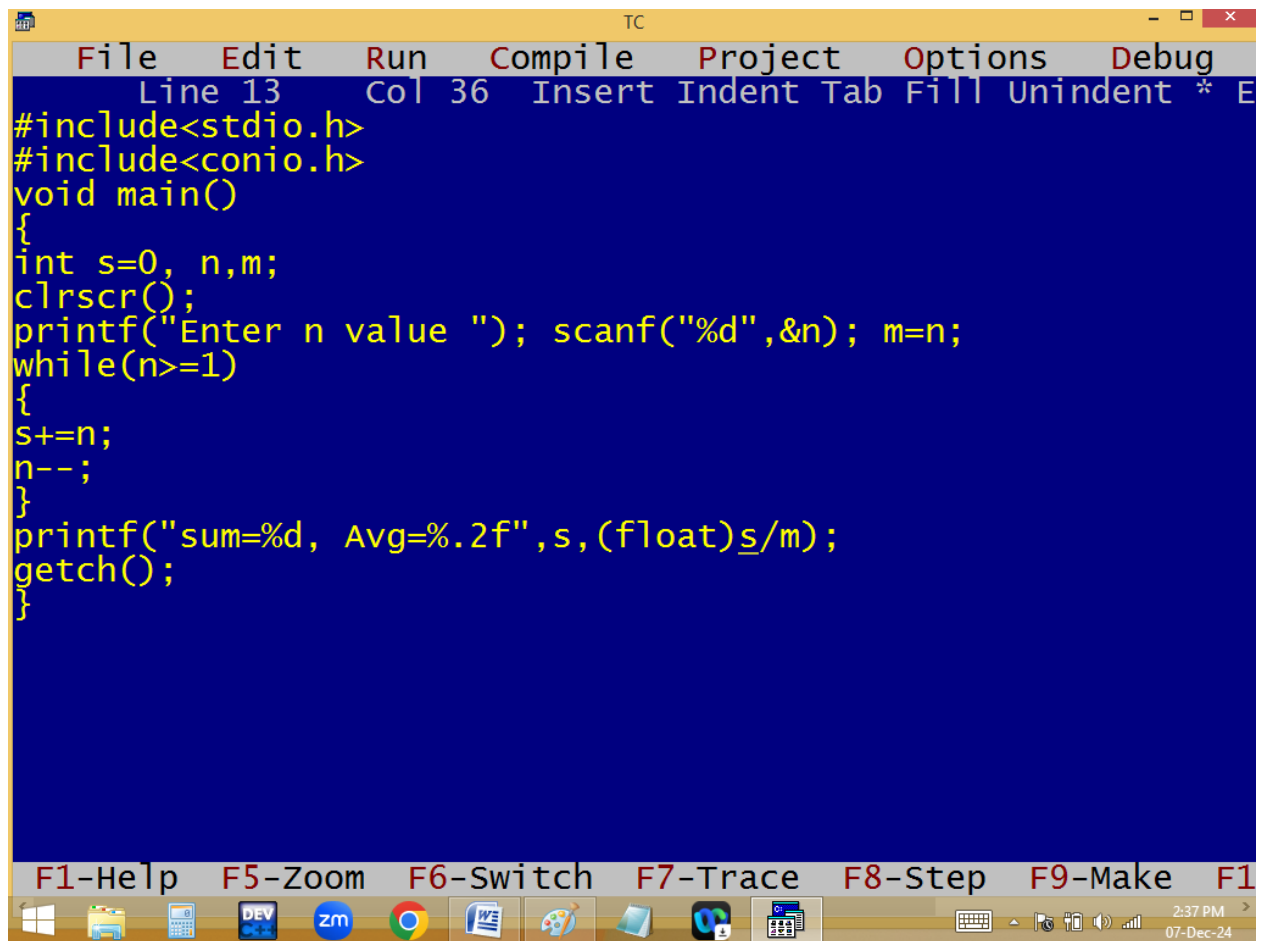
```
#include<stdio.h>
#include<conio.h>
void main()
{
int es=0, os=0, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)
{
if(n%2==0)es+=n; else os+=n;
n--;
}
printf("Even sum=%d, Odd sum=%d",es, os);_
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "2:33 PM" and "07-Dec-24".





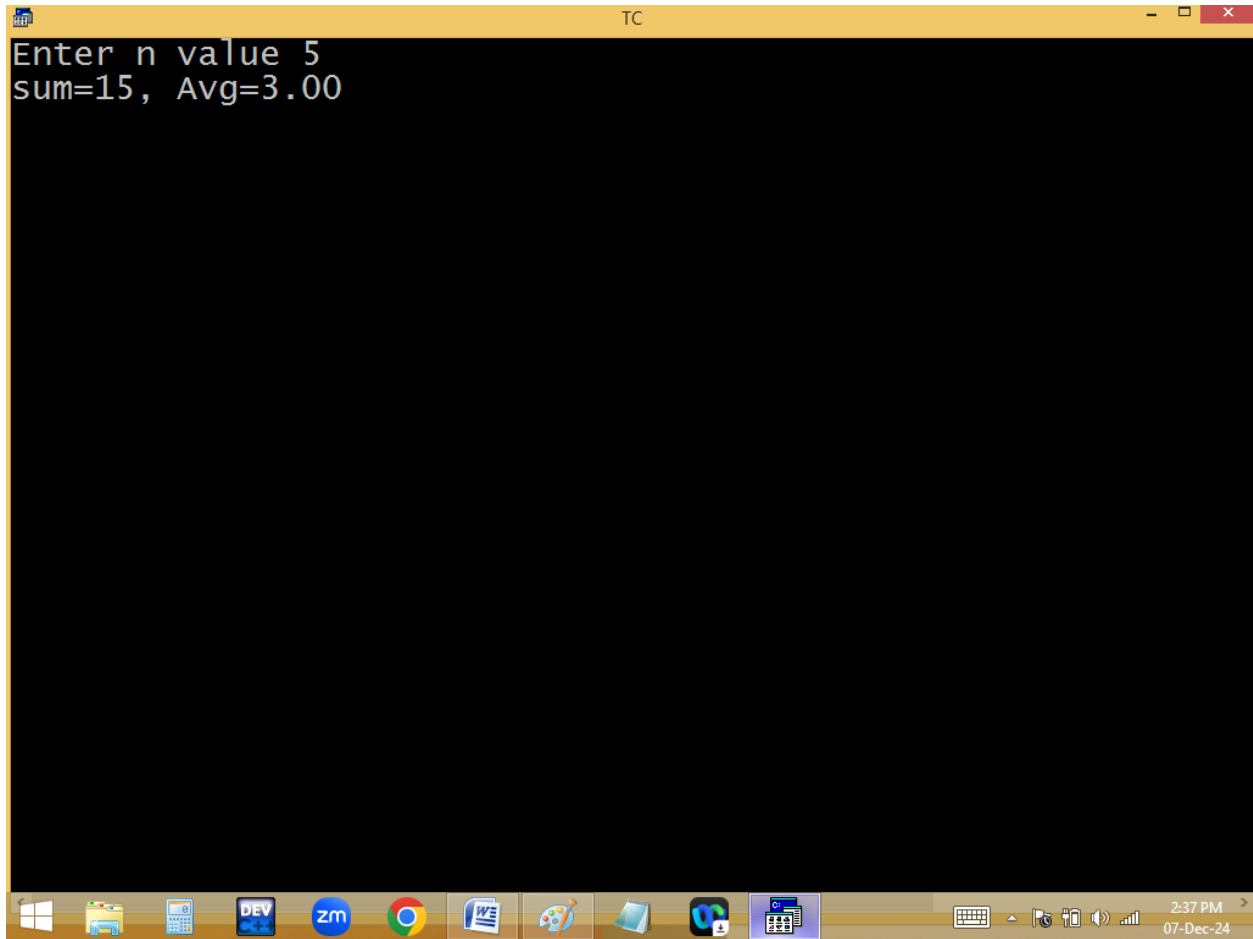
Find 1..n no's sum and avg [mean]:

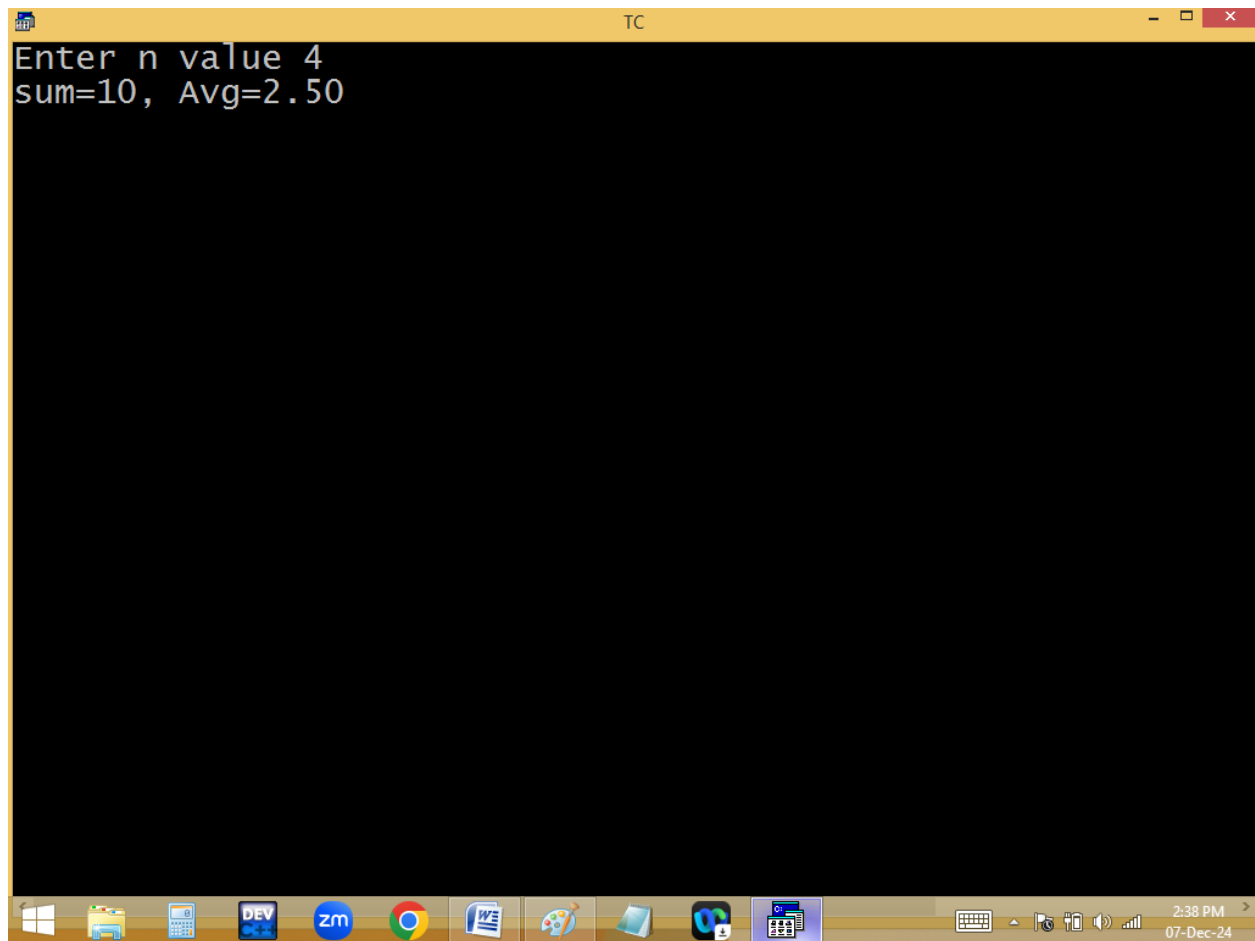


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 13 Col 36 Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes `<stdio.h>` and `<conio.h>`, defines a `main()` function, declares variables `s`, `n`, and `m`, clears the screen with `clrscr()`, prompts the user to enter a value for `n` using `scanf`, and then enters a `while` loop to calculate the sum `s` and the average `s/m`. The program ends with `printf` to display the results and `getch()` to pause the execution.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int s=0, n,m;
clrscr();
printf("Enter n value "); scanf("%d",&n); m=n;
while(n>=1)
{
s+=n;
n--;
}
printf("sum=%d, Avg=%.2f",s,(float)s/m);
getch();
}
```

The bottom of the window features a toolbar with icons for various functions and a status bar at the very bottom showing the time as 2:37 PM and the date as 07-Dec-24.

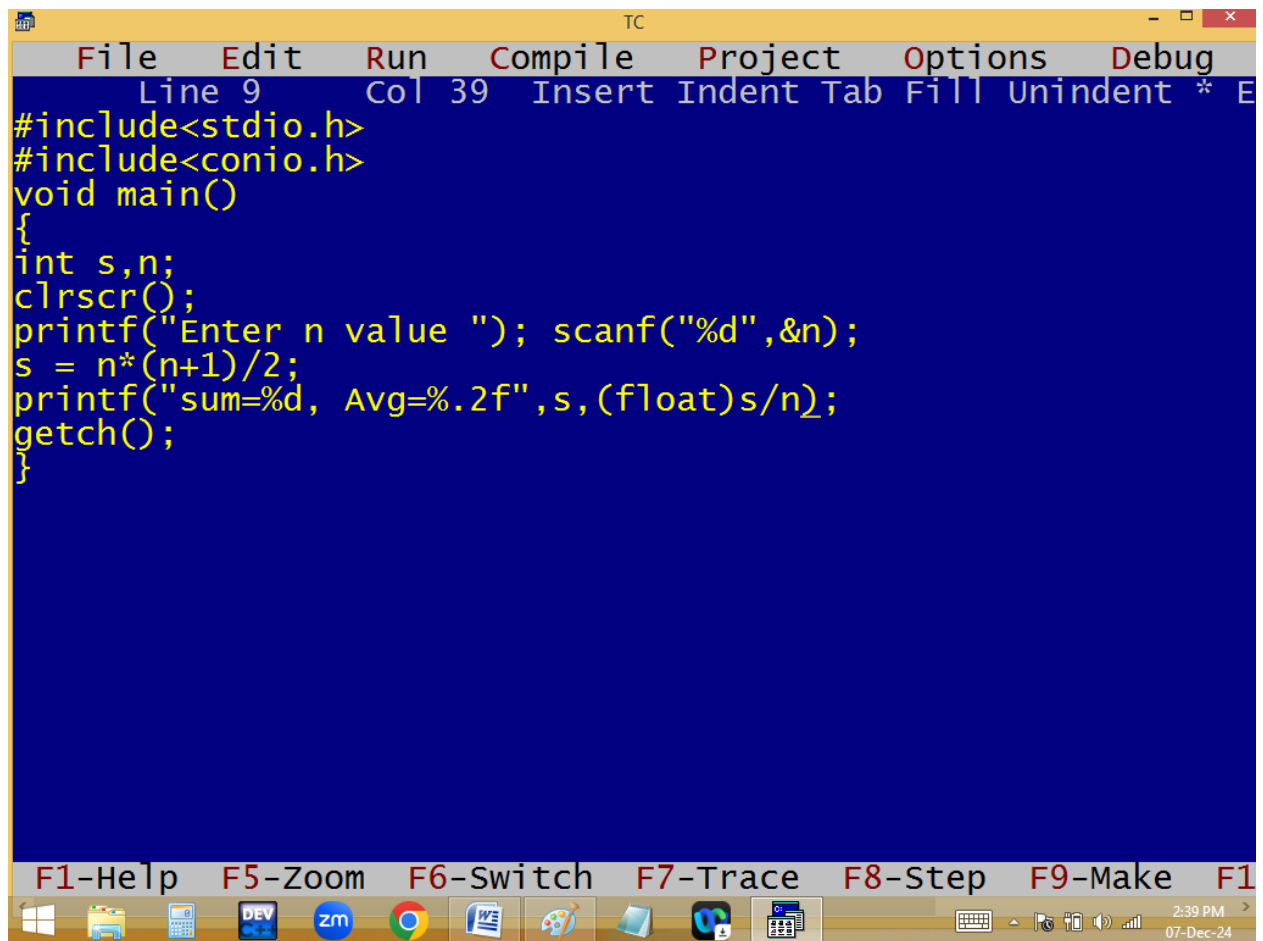




The image shows a screenshot of a Turbo C++ (TC) window. The window title bar is yellow and contains the text "TC". The main area of the window is black and displays the following text in white: "Enter n value 4" and "sum=10, Avg=2.50". The Windows taskbar is visible at the bottom, showing various application icons including File Explorer, Calculator, DEV, zm, Chrome, Word, and others. The system tray on the right shows the time as 2:38 PM and the date as 07-Dec-24.

```
Enter n value 4
sum=10, Avg=2.50
```

Without using loop / goto label:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 9", "Col 39", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int s,n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
s = n*(n+1)/2;
printf("sum=%d, Avg=%.2f",s,(float)s/n);
getch();
}
```

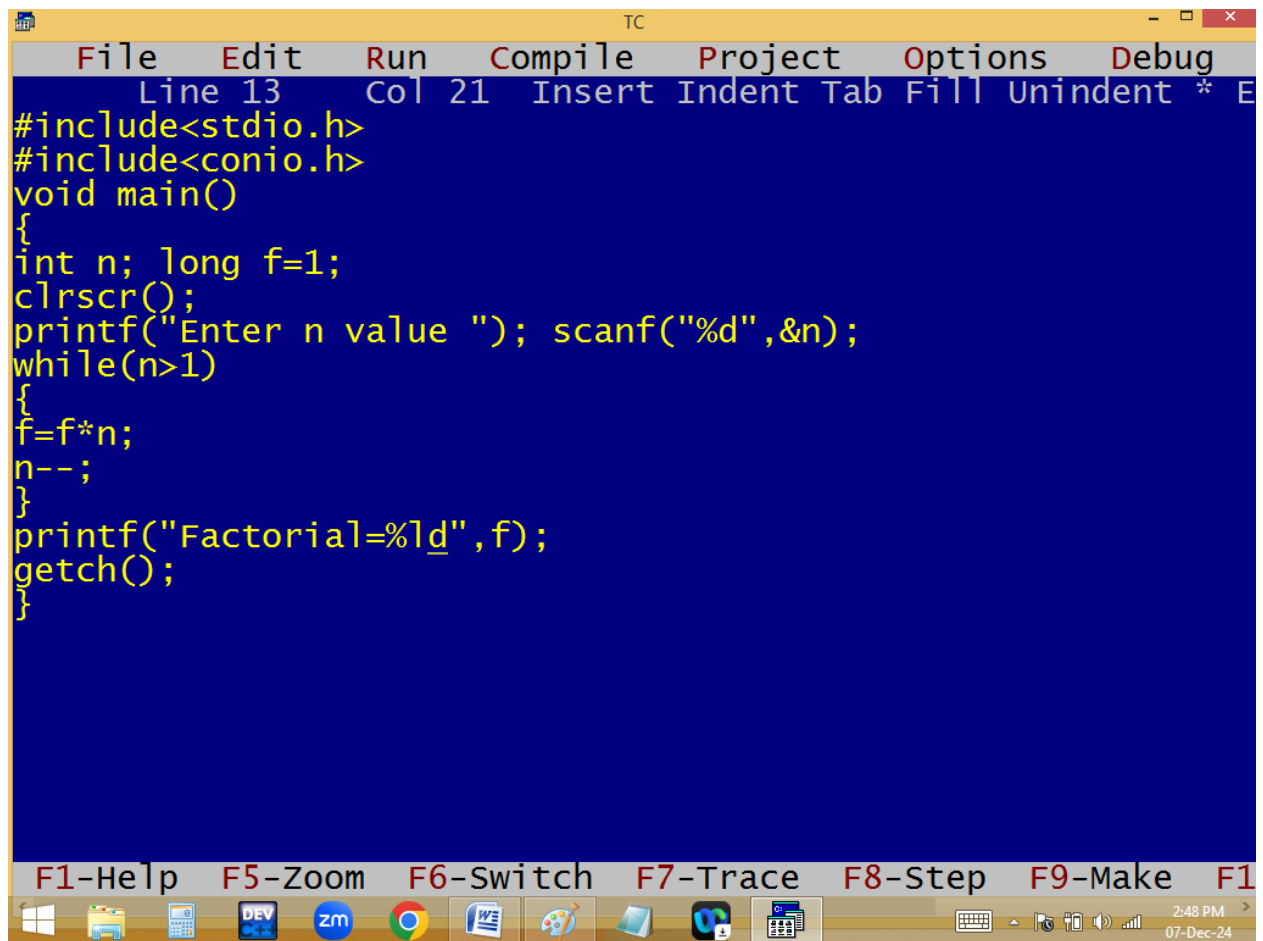
At the bottom of the window, there is a toolbar with icons for various functions and a status bar. The status bar shows the time "2:39 PM" and the date "07-Dec-24".

```
TC
Enter n value 6
sum=21, Avg=3.50
```

$$\frac{n}{4}$$

$$s = n * (n+1) / 2 ==> 4 * 5 / 2 = 10$$

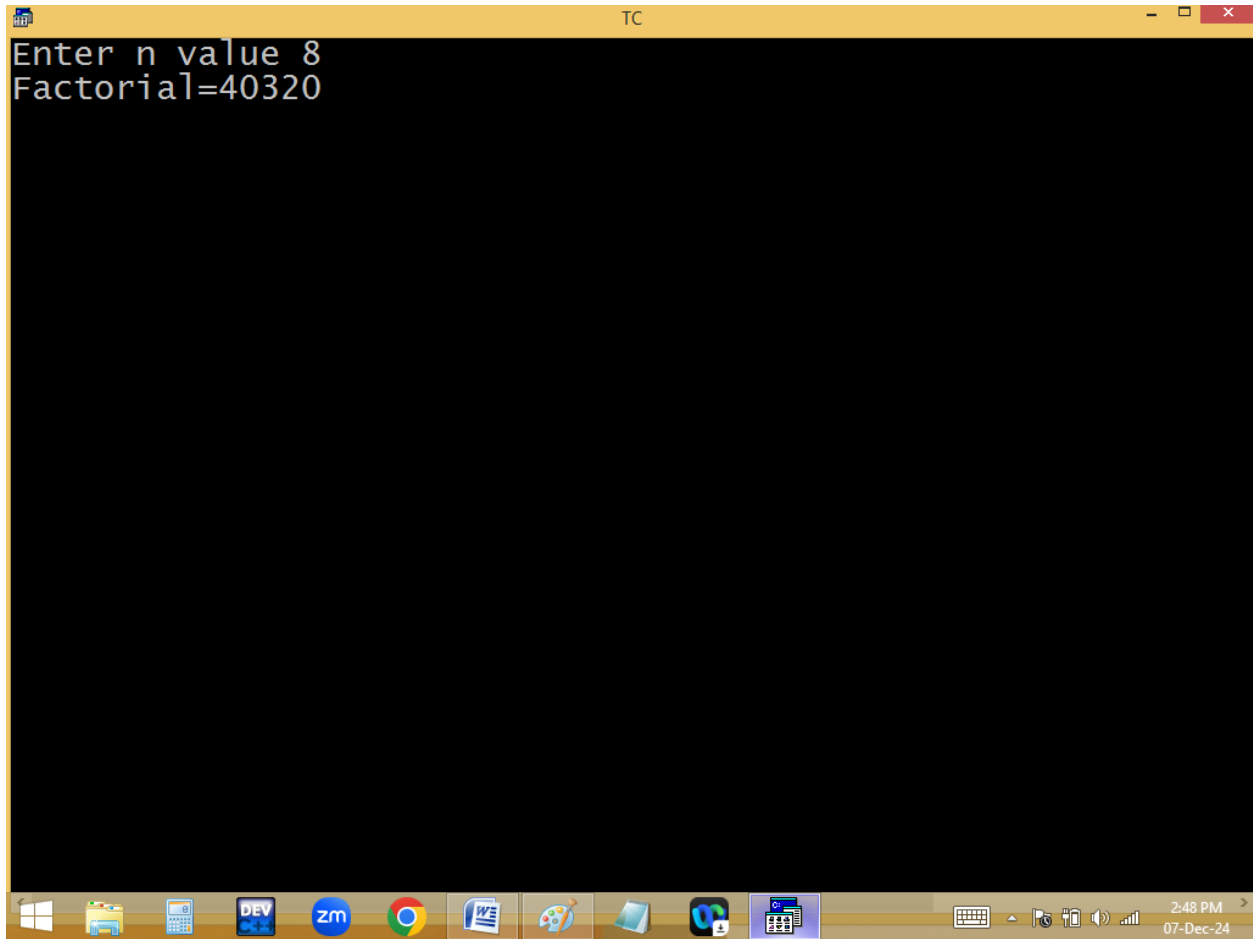
Finding factorial of given no



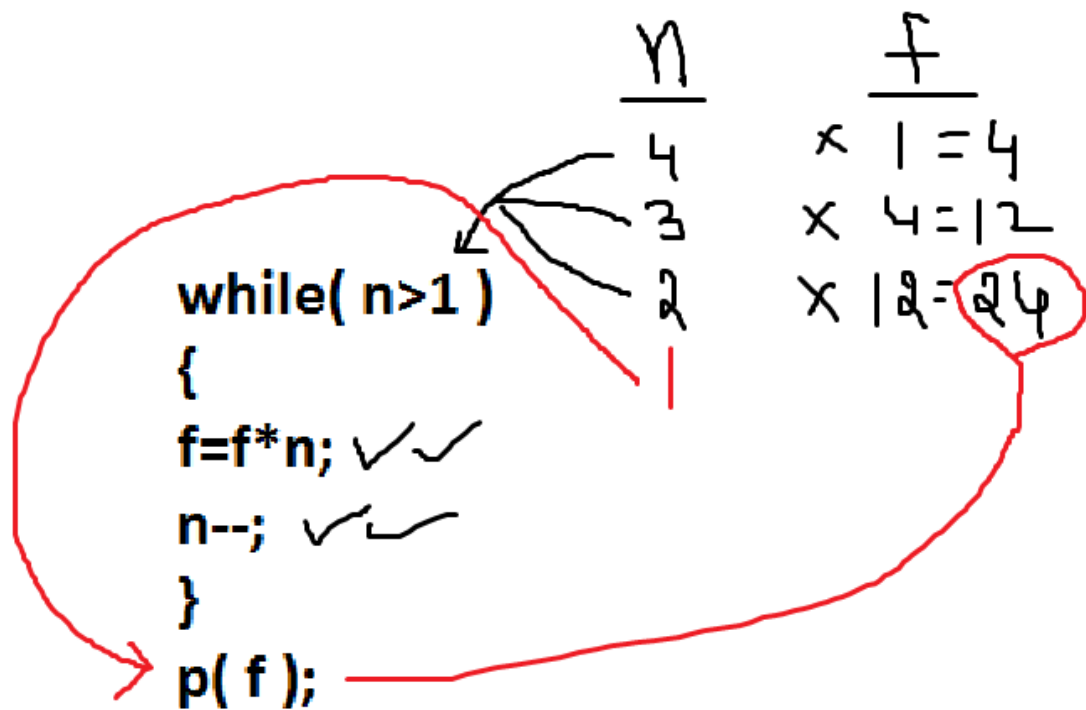
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 13", "Col 21", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code for calculating a factorial:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n; long f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial=%ld",f);
getch();
}
```

At the bottom of the IDE window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the toolbar is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and a folder icon. The system tray on the right shows the time "2:48 PM" and the date "07-Dec-24".

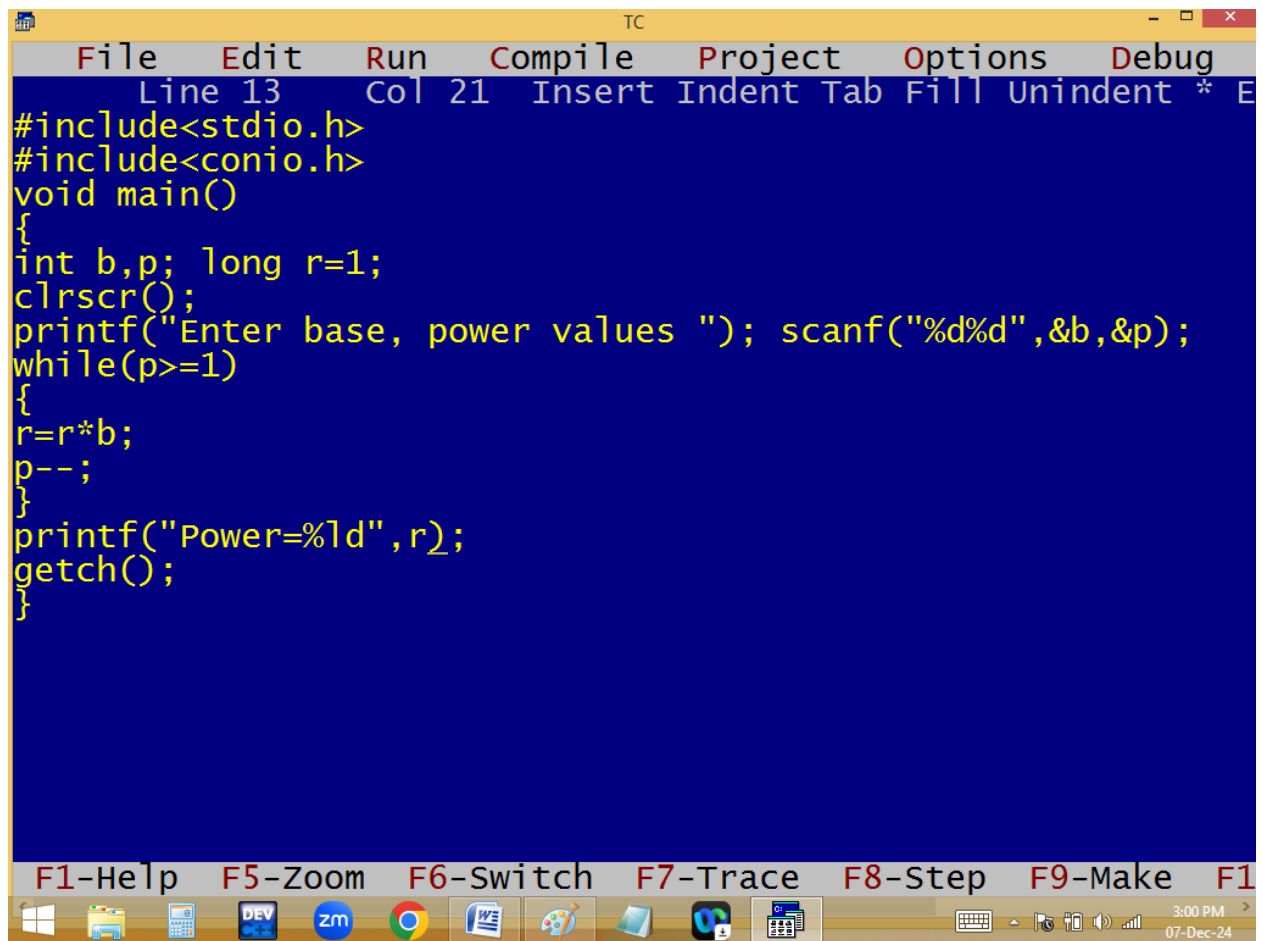


```
Enter n value 10
Factorial=3628800_
```



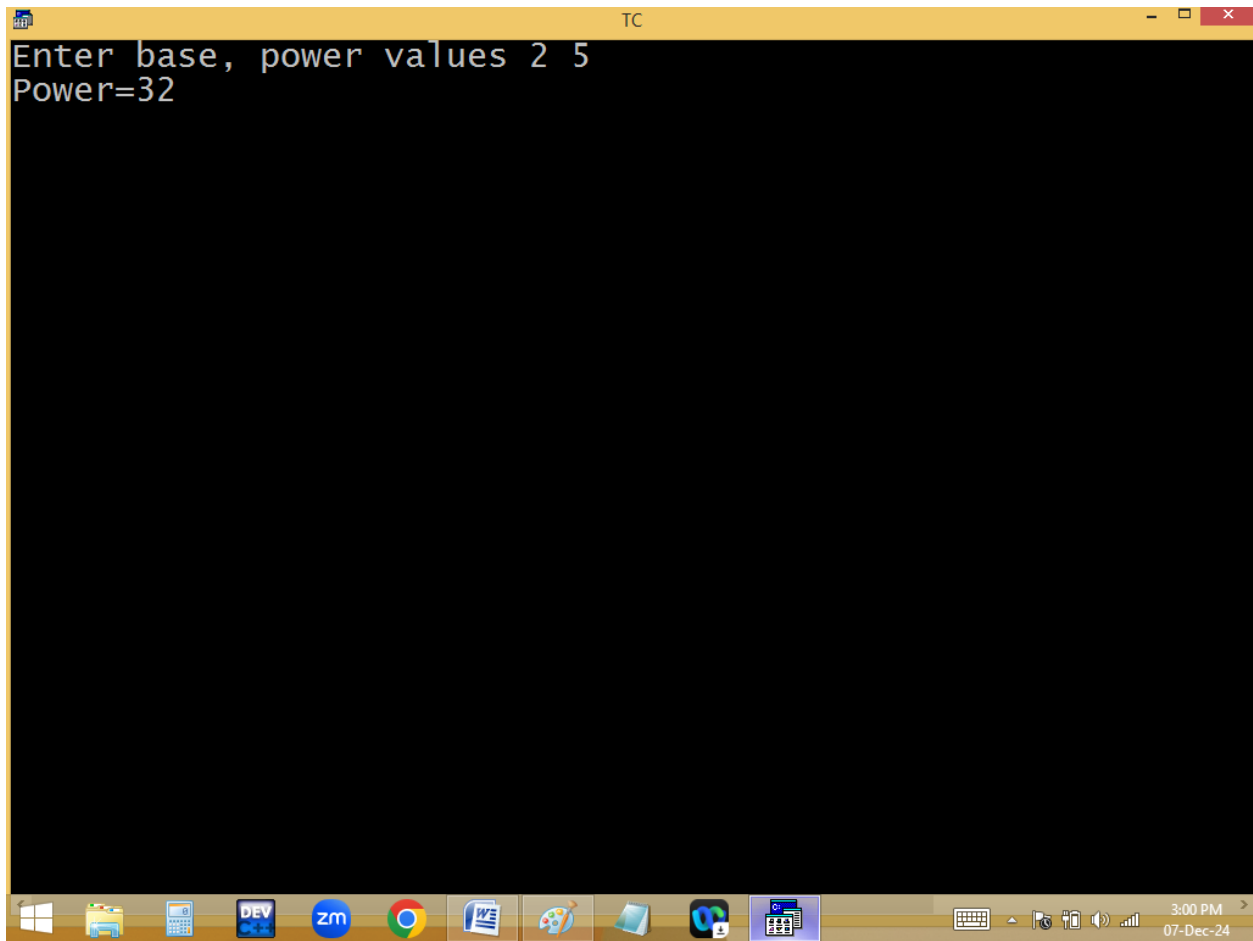
Finding power using user defined program?

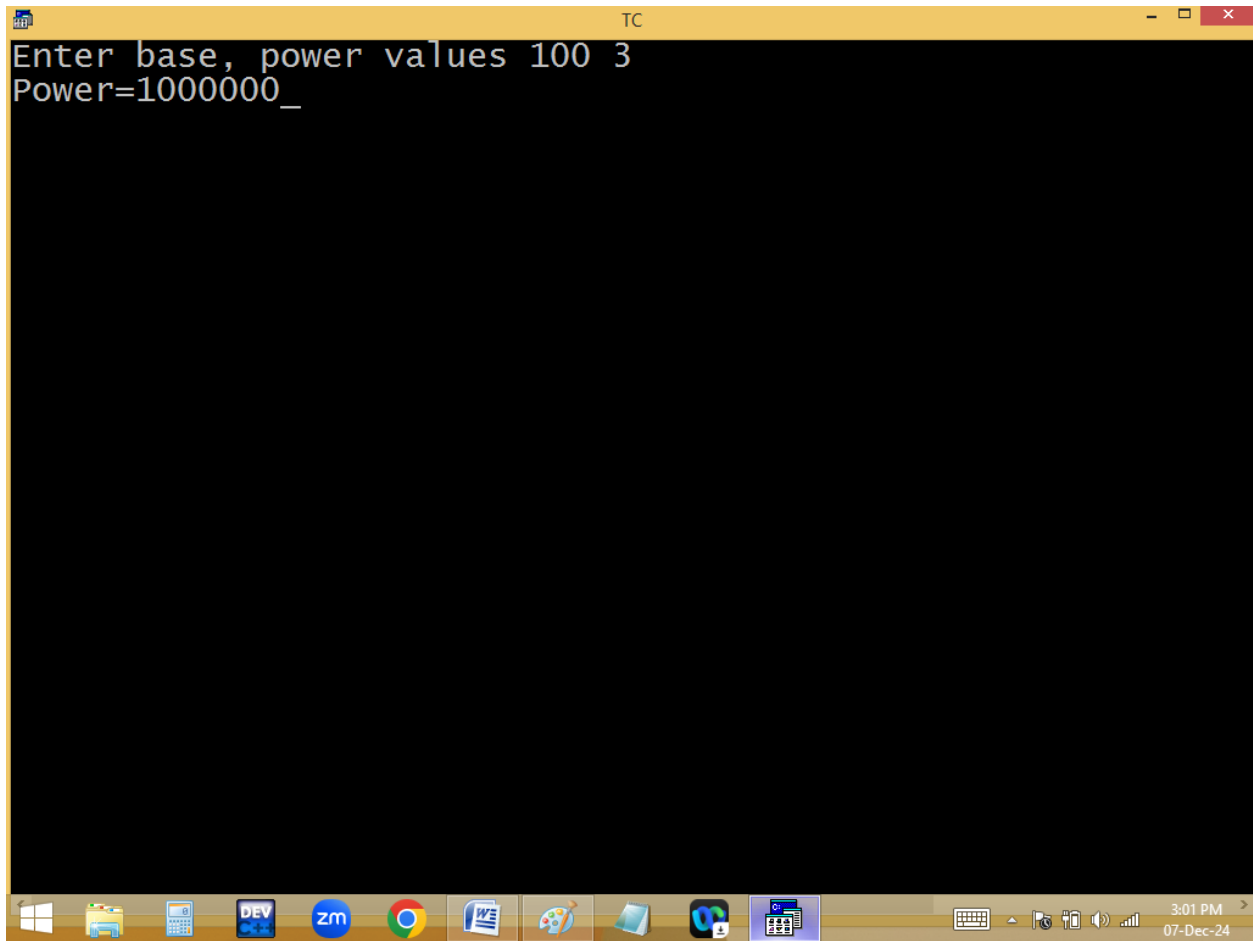
$$2^5 = 32$$

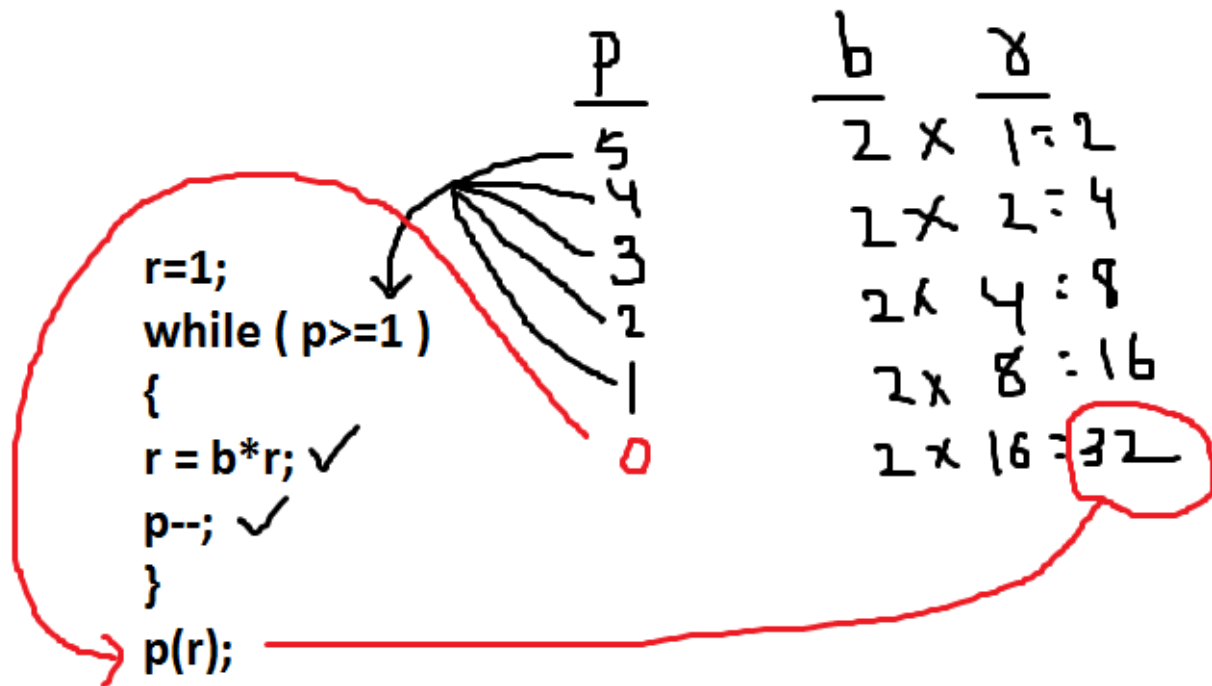


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 13", "Col 21", and "Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, it declares `int b, p;` and `long r=1;`, calls `clrscr()`, and prompts the user to enter base and power values using `scanf("%d%d",&b,&p);`. A `while(p>=1)` loop calculates the power by multiplying `r` by `b` and decrementing `p`. The result is printed using `printf("Power=%ld",r);`, and the program ends with `getch();`. The bottom status bar shows function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock shows "3:00 PM" and "07-Dec-24".

```
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p; long r=1;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p>=1)
{
r=r*b;
p--;
}
printf("Power=%ld",r);
getch();
}
```







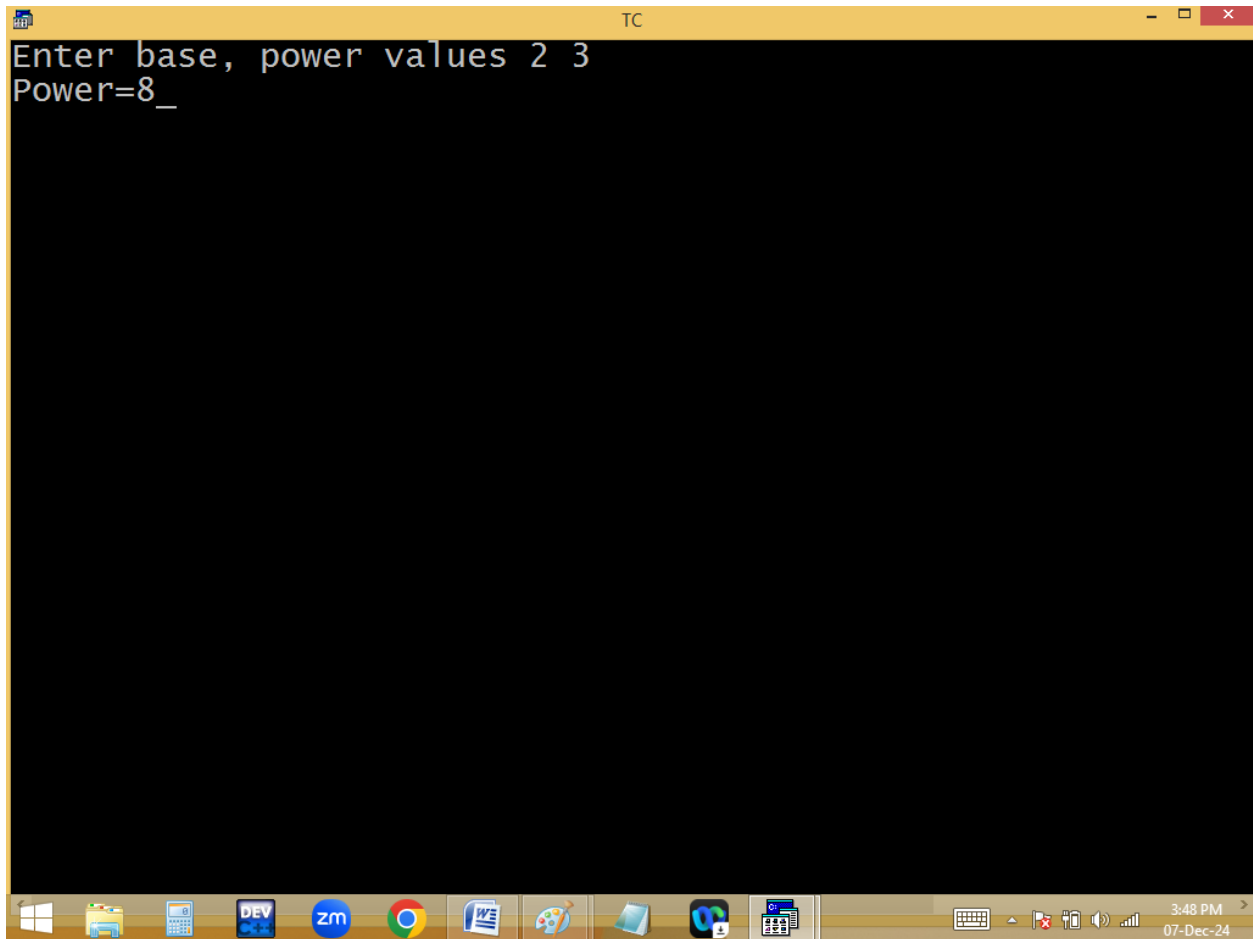
$$2^5 \Rightarrow \underline{2*2*2*2*2=32}$$

Using user defined function:

```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void power(); /* fun declaration */
void main()
{
clrscr();
power(); /* fun calling */
getch();
}
void power() /* fun definition */
{
int b,p; long r=1;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p>=1)
{
r=r*b;
p--;
}
printf("Power=%ld",r);
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

3:47 PM 07-Dec-24



The image shows a screenshot of a Turbo C++ (TC) window. The window has a yellow title bar with the text "TC" and standard minimize, maximize, and close buttons. The main area is black with white text. The text reads: "Enter base, power values 2 3" followed by "Power=8_". The window is running on a Windows operating system, as evidenced by the taskbar at the bottom. The taskbar contains several icons: Windows Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Microsoft Word, Paint, a folder icon, a game icon, and a calendar icon. The system tray on the right shows the time as 3:48 PM and the date as 07-Dec-24.

What is a fun [happy] + action [moments]?

Function is a small program is used to do a particular task [work].

Function is a sub program / sub routine / procedure / module / structure

It is a reusable code component

It is a small program within another program

It is a self contained block.

Udf is divided into 3 parts.

1.Function declaration / proto type: conducted before / within main()

Eg: void sum();

2.Function calling: should be within main()

Eg: sum();

3.Function definition:

Eg:

```
void sum()
```

```
{
```

```
Int a=10, b=20;
```

```
Printf("Sum=%d",a+b);
```

```
}
```

Turbo C++ IDE

File Edit Search Run Compile Debug Project Optio

NONAME00.CPP

```
#include<stdio.h>
#include<conio.h>
//void sum(); // fun declaration
void sum() //callie, fun definition
{
int a=10, b=20;
printf("Sum=%d\n",a+b);
}

void main() // caller
{
clrscr();
sum(); // fun calling
getch();
}
```

14:30

F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile

3:56 PM 07-Dec-24

